

Partho Bala

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EXPERIENCE

MACHINE LEARNING ENGINEER

JUNE 2022 - PRESENT

Technonext Software Ltd.

- Developed an ASR based travel assistant **Chatbot** and successfully integrated in to a mobile app. [link](#)
- Developed and deployed an end-to-end ML pipeline for personalized **restaurant recommendations** for food delivery service, including data preprocessing, candidate generation, and ranking.
- Built an LLM-driven product tagging pipeline that analyzes product titles to generate accurate tags, with validation mechanisms to reduce hallucinations and ensure tag quality
- Implemented an image-based category classification pipeline using CLIP embeddings to support **visual search** and improve product categorization.
- Designed and deployed **OCR** pipeline for English handwritten documents, achieving over 86% extraction accuracy across 5+ invoice formats..
- Implemented **NER** based address extractor for online courier service.
- Improved the accuracy of a **face verification** system by integrating new functionalities
- Developed a web scraper to monitor real-time airline ticket prices, enabling dynamic price tracking and insights for optimized pricing strategies

Junior Software Engineer

JAN 2022 – MAY 2022

Swosti Limited

- Responsible for maintaining, implementing critical bug fix and supporting the existing application.
- Design and develop user friendly report using Birt and Jasper Report.

EDUCATION

BSC IN COMPUTER SCIENCE AND ENGINEERING

JAN 2016 – JUNE 2021

Bangabandhu Sheikh Mujibur Rahman Science and Technology University

PUBLICATION

ANALYZING EMPLOYEE RETENTION FACTORS USING MACHINE LEARNING

Moshiur Rahman, Md Rashedul Islam, **Partho Bala**, Abdus Sattar, IEEE 2024

DOI: <https://doi.org/10.1109/ICAECT60202.2024.10469009>

Skills

Programming Language : Python,

Backend: FastAPI

Machine Learning: Rasa, LangChain, Tensorflow, Pytorch, Pandas, Numpy, Spacy, HuggingFace

Automation: Selenium, BeautifulSoup

Database: Redis, PostgreSQL.

Visualization: Matplotlib, Seaborn

Other: Git, Linux, Docker , Jira

PERSONAL PROJECTS

- **Bengali Clickbait Detection Using NLP**

A NLP application which can detect whether the news is clickbait or not by performing Sentiment analysis on news headlines.

- Build a dataset by collecting different types of clickbait and authentic news headlines.
- Perform different types of ML models and build neural networks to get score upto 81%.

Technology Used: Python, Scikit-learn, Gensim, Keras.

- **Country Classifier** ([Link](#))

This project classifies a country based on a given text. It utilizes the BERT model to understand the context of the text and predict the country.

Technology Used: Python, Pandas, Pytorch, Huggingface, Gradio

- **Auction Site** ([Live](#) , [Code](#))

A simple e-commerce site like ebay where user can add products for auction. User can also bid items before the end of auction. After ending of an auction a winner of that auction is announced.

Technology Used: Django, Postgresql, Bootstrap.

- **Distance Learning** ([Code](#))

An e-learning platform where instructor can add course materials and student can enroll course Online. A group chat system also added to discuss about course topics.

- For chat server use **Redis** for enabling channel layer, implement **Websocket** and use **Channels** to make server fully asynchronous.
- Build a RESTful **API** with JSON serialization.

Technology Used: Django, JQuery, Redis, Html/CSS.

- **Personal Blog** ([Code](#))

A simple blog application where viewers can search, comment on blog post and also can subscribe RSS feed.

Technology Used: Django, Postgresql, Bootstrap.

Extra-Curricular Activities

- Contestant at Nasa Space Apps Challenge 2019 (Regional Finalist).
 - Academic Member and Volunteer of National High School Programming Contest(NHSPC)- 2016-2017.
 - Contestant at UGB ICT Carnival
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